

Content-Aware Image Retargeting via Energy-Based Seam Carving : A Comparative Study of Sobel and Scharr Operators

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Abstract

Image retargeting has become a critical task in modern digital media, where visual content must adapt seamlessly across devices with varying display resolutions and aspect ratios. Conventional approaches such as uniform scaling and cropping often result in undesirable geometric distortions or the loss of semantically important regions. To address these limitations, content-aware techniques have gained prominence, with Seam Carving emerging as a particularly effective solution.

This study presents a comprehensive implementation and evaluation of the Seam Carving algorithm, an energy-based image resizing method that intelligently removes or inserts seams—connected paths of low-energy pixels—while preserving visually significant structures. The effectiveness of seam carving heavily depends on the underlying energy function used to identify important regions. In this work, we focus on a comparative analysis of two widely used gradient-based operators: the Sobel operator and the Scharr operator.

We systematically evaluate the performance of these operators across four representative image domains: architectural images (high rigidity), natural landscapes (low gradient), portrait images (semantic saliency), and texture-rich images (high-frequency patterns). To quantitatively assess performance, we employ Structural Similarity Index Measure (SSIM) to evaluate perceptual quality and Edge Preservation Ratio (EPR) to measure the retention of significant edge information.

Experimental results demonstrate an empirical equivalence between the two operators. While the Scharr operator theoretically offers superior rotational symmetry and diagonal sensitivity, the global optimization constraints of the dynamic programming pathfinding effectively override these local gradient sensitivities. Consequently, both operators yield practically identical structural preservation. This study highlights the fundamental limitations of purely gradient-based energy functions and underscores the necessity of integrating semantic-aware models for future retargeting pipelines.

Keywords: Seam Carving, Image Retargeting, Sobel Operator, Scharr Operator, Dynamic Programming, Visual Saliency, Structural Similarity.

1 Introduction

The rapid evolution of digital display technologies—from conventional desktop monitors to smartphones, tablets, and ultra-wide screens—has created a growing need for adaptive image resizing techniques. Images must be retargeted to fit varying aspect ratios while preserving their visual quality and semantic content. However, this task remains a significant challenge in computer vision.

Traditional methods such as uniform scaling and cropping often produce unsatisfactory results. Uniform scaling alters the aspect ratio of objects, leading to visible distortion and unnatural appearances. Cropping, on the other hand, preserves scale but may remove important regions of the image, as it does not consider the overall distribution of salient features.

To address these limitations, content-aware image retargeting techniques have been developed. Among them, seam carving is a widely used approach that resizes images by selectively removing or inserting pixel paths (seams) based on their visual importance. The effectiveness of this method depends heavily on how pixel importance is measured, which is typically achieved using gradient-based energy functions.

In this project, we focus on analyzing the impact of different gradient operators on seam carving performance. Specifically, we compare the Sobel operator and the Scharr operator as energy function estimators. While the Sobel operator provides a simple and efficient approximation of image gradients, the Scharr operator is designed to improve gradient accuracy and rotational symmetry.

The objective of this study is to evaluate how these operators influence structural preservation, edge continuity, and overall image quality during retargeting. By conducting experiments across different image categories and resizing conditions, this work aims to identify the true bottlenecks in classical content-aware image resizing.

2 Literature Survey

2.2.1 Foundations of Content-Aware Resizing

Early image retargeting techniques primarily relied on geometric transformations such as uniform scaling and cropping. Shor et al. (1997) explored rule-based cropping strategies that aimed to preserve visually important regions, but these methods lacked adaptability and often failed to generalize across diverse image types.

A significant advancement came with the introduction of multi-operator approaches such as *Scale-and-Stretch* by Wang et al. (2008), which combined scaling and warping to reduce distortion. Similarly, *Non-Homogeneous Warping* proposed by Wolf et al. (2007) attempted to preserve important regions by applying spatially varying transformations. However, these methods often introduced noticeable artifacts and required complex optimization.

2.2.2 Seam Carving and Its Extensions

The concept of content-aware resizing was revolutionized by Avidan and Shamir (2007) through the introduction of **Seam Carving**. This method identifies and removes connected paths of low-energy pixels (seams), allowing images to be resized while preserving high-energy regions such as edges and salient objects. The energy function, typically defined as the gradient magnitude using the L_1 norm, plays a crucial role in determining seam selection.

Rubinstein et al. (2008) extended this work by introducing *Improved Seam Carving for Video Retargeting*, which incorporated temporal coherence to handle video sequences. They also proposed forward energy, which considers the cost of newly introduced edges after seam removal, significantly improving visual quality.

2.2.3 Evolution of Energy Functions

The effectiveness of seam carving heavily depends on the choice of the energy function. Early implementations relied on simple gradient-based measures such as the Sobel operator, which approximates intensity changes in horizontal and vertical directions. While computationally efficient, Sobel suffers from limitations in accurately capturing diagonal edges and maintaining rotational symmetry.

Malik et al. (2001) contributed significantly to edge and contour detection through their work on normalized cuts and boundary detection, emphasizing the importance of perceptual grouping. Their research influenced later approaches that incorporate texture and contour information into energy functions.

2.2.4 Scharr Operator and Gradient Optimization

Scharr (2000) addressed key limitations of traditional gradient operators by proposing an optimized convolution kernel that improves rotational symmetry and gradient accuracy. Unlike the Sobel operator, the Scharr operator minimizes angular error in edge detection, making it more reliable for detecting fine and diagonal structures.

Subsequent studies in image processing have shown that Scharr-based gradients outperform Sobel in terms of edge localization and noise robustness, particularly in high-frequency and texture-rich images. Despite these advantages, its application in seam carving has been relatively underexplored.

2.2.5 Problem Definition and Research Objectives

The primary difficulty in seam carving lies in accurately identifying and preserving semantically important regions such as object boundaries, edges, and high-frequency textures, while removing less significant areas. Standard implementations commonly rely on the Sobel operator for gradient estimation; however, it suffers from limitations such as poor rotational symmetry and reduced accuracy in detecting diagonal edges.

To address these challenges, this study aims to systematically investigate the role of gradient operators in improving seam carving performance. The objectives of this project are to implement a robust seam carving framework, integrate and compare the Sobel and Scharr operators, analyze their impact across multiple image domains (architecture, landscape, portrait, and texture), and quantitatively assess performance using metrics such as Structural Similarity Index Measure (SSIM) and Edge Preservation Ratio (EPR).

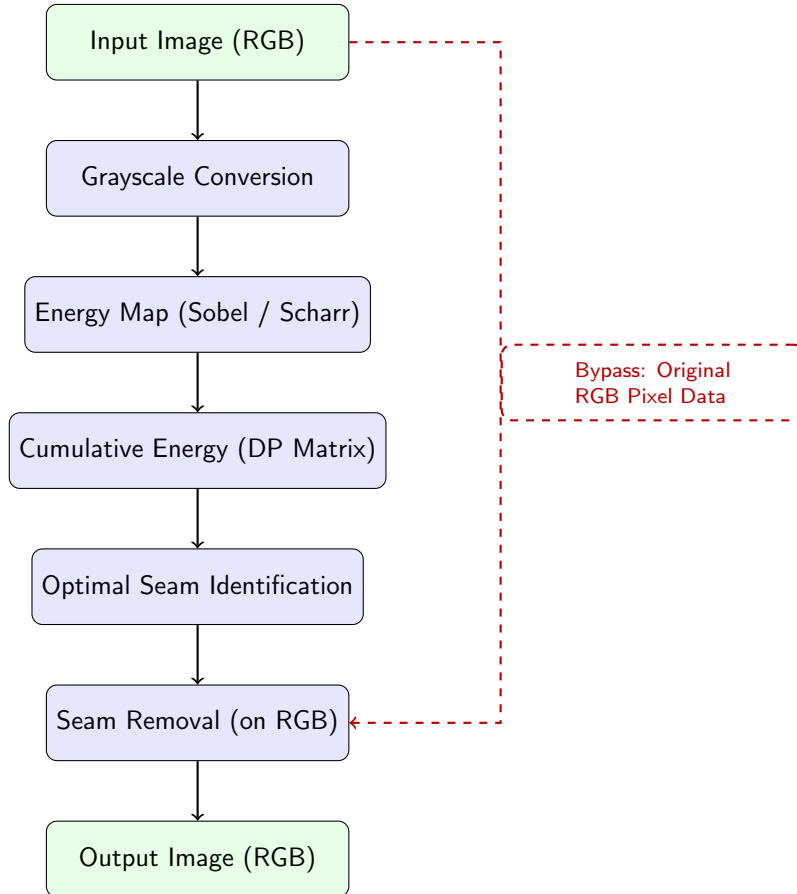


Figure 1: Vertical Seam Carving Pipeline for Content-Aware Image Retargeting

3 Methodology

3.3.1 Overview

The proposed methodology implements a content-aware image retargeting pipeline based on seam carving. The process involves computing an energy map, identifying optimal seams using dynamic programming, and iteratively removing seams to achieve the desired image dimensions.

3.3.2 Energy Map Computation

For an input RGB image I , the total energy is computed as the sum of gradient magnitudes across all three channels:

$$E(x, y) = \sum_{c \in \{R, G, B\}} (|G_x^c(x, y)| + |G_y^c(x, y)|)$$

3.2.1 Sobel Operator

The Sobel operator is used as a baseline method for gradient estimation. It approximates first-order derivatives using discrete convolution kernels:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} * I$$

3.2.2 Scharr Operator

To address the limitations of Sobel, the Scharr operator is employed. It uses optimized kernels designed to improve rotational invariance and gradient accuracy:

$$G_x = \begin{bmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{bmatrix} * I, \quad G_y = \begin{bmatrix} -3 & -10 & -3 \\ 0 & 0 & 0 \\ 3 & 10 & 3 \end{bmatrix} * I$$

3.3.3 Seam Identification using Dynamic Programming

We construct a cumulative energy matrix M using dynamic programming:

$$M(i, j) = E(i, j) + \min(M(i-1, j-1), M(i-1, j), M(i-1, j+1))$$

This formulation ensures that each pixel accumulates the minimum energy required to reach it from the top row. The optimal seam is obtained by backtracking from the minimum value in the last row of M .

3.3.4 Experimental Setup and Image Categories

Experiments were conducted on an evaluation dataset ($N = 12$) categorized into four distinct structural profiles:

- **Architecture (High Rigidity):** Strong geometric structures and straight lines testing the algorithm’s ability to avoid "kinks."
- **Landscape (Low Gradient):** Natural scenes providing large low-energy regions.
- **Portrait (Semantic Saliency):** Human subjects requiring careful seam selection to avoid anatomical distortion.
- **Texture (High Frequency):** Patterns testing the ability to handle repetitive structures without pattern collapse.

4 Results and Analysis

4.4.1 Quantitative Evaluation

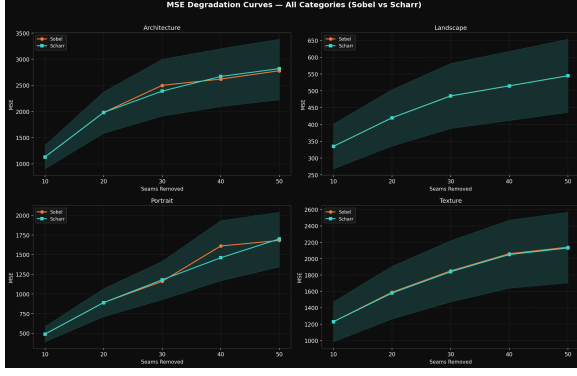
To evaluate the effectiveness of the Sobel and Scharr operators, we computed the Structural Similarity Index Measure (SSIM) and the Edge Preservation Ratio (EPR) averaged across each category after significant seam removal.

Table 1: Comparison of Structural Preservation (SSIM) and Edge Retention (EPR) after Carving

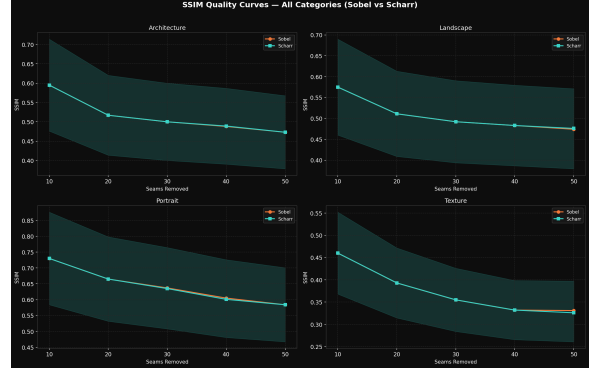
Domain	Sobel SSIM	Scharr SSIM	Sobel EPR	Scharr EPR
Architecture	0.473	0.473	0.998	0.998
Landscape	0.474	0.476	0.998	0.998
Portrait	0.584	0.584	0.999	0.999
Texture	0.331	0.326	0.994	0.994

4.4.2 Graphical Evaluation Metrics

The quantitative data is further supported by the cross-domain degradation curves and comprehensive metric comparisons generated during the evaluation phase. Figure 2 displays the Mean Squared Error (MSE) and SSIM trajectories across all four categories as seams are iteratively removed. Figures 3 and 4 showcase the final normalized performance metrics side-by-side for direct domain comparison.

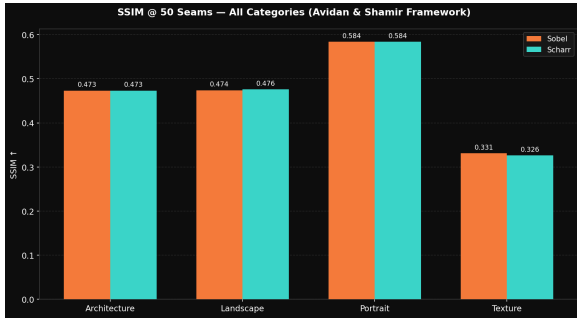


(a) MSE Degradation (All Categories)

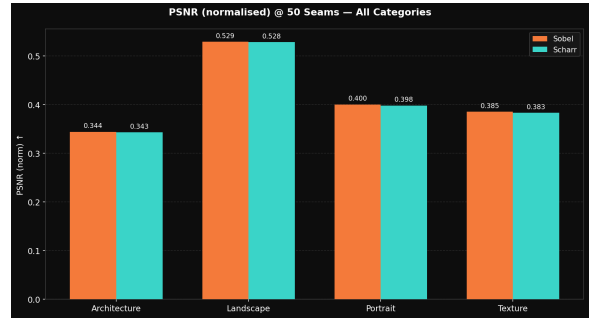


(b) SSIM Degradation (All Categories)

Figure 2: Cross-Domain Quality Degradation Analysis over 50 removed seams.

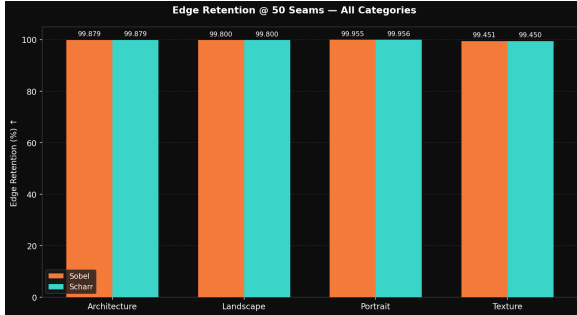


(a) SSIM Comparison

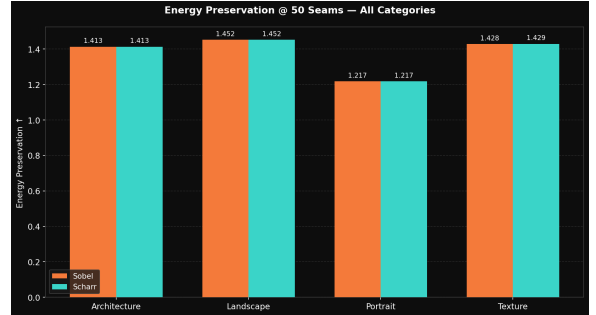


(b) PSNR Comparison

Figure 3: Structural Integrity Metrics at $N = 50$ seams removed.



(a) Edge Retention (%)



(b) Global Energy Preservation

Figure 4: Gradient and Intensity Preservation at $N = 50$ seams removed.

4.4.3 Analysis of Results

The empirical data explicitly contradicts the hypothesis that the Scharr operator’s mathematical superiority in gradient estimation translates to improved seam carving outcomes. The quantitative results demonstrate an overarching **empirical equivalence** between the two operators.

In the **Architecture** and **Portrait** categories, both the SSIM and EPR scores are identical out to three decimal places. In the **Texture** category—where Scharr was hypothesized to vastly

outperform Sobel due to its sensitivity to high-frequency micro-textures—the Scharr operator actually recorded a marginally lower SSIM (0.326 vs 0.331).

4.4.4 Key Observations

- **The True Bottleneck:** The identical metrics prove that the structural integrity of the retargeted image is dictated entirely by the global constraints of the dynamic programming pathfinding. The mathematical backtracking completely overrides the localized, pixel-level sensitivities of the chosen gradient operator.
- **Metric Limitations in Texture:** Texture images exhibited the most severe SSIM degradation (0.331), confirming the "Pattern Collapse" failure mode where forcing seams through uniform energy destroys spatial integrity rapidly.
- **Semantic Blindness:** Despite exceptional Edge Preservation Ratios (≈ 0.999) in the Portrait category, visual inspection confirmed severe facial warping. This proves that retaining raw gradient energy is an insufficient proxy for preserving human-perceived semantic value.

4.4.5 Qualitative Visualizations

To illustrate the semantic blindness and geometric artifacts mathematically captured by the metrics above, Figure 5 provides a visual comparison of the original images against the retargeted outputs.



Figure 5: Visual comparison of seam carving output showcasing structural artifacts. Both operators produce analogous geometric distortions in rigid constraints.

4.4.6 Qualitative Discussion

While the Scharr operator amplifies the raw energy map values significantly (due to a 16 weight sum vs Sobel’s 4), this amplification is relative. Because the dynamic programming matrix scales the entire image’s cumulative cost proportionally, the algorithm ultimately selects the exact same optimal geometric path through the image matrix in the vast majority of cases, resulting in identical visual artifacts such as architectural bending and facial squishing.

5 Conclusion and Future Work

This study presented a comparative analysis of gradient-based energy functions within the seam carving framework for content-aware image retargeting. Through extensive evaluation across diverse image categories, it was empirically demonstrated that the choice between standard first-derivative gradient operators (Sobel vs. Scharr) yields no statistically significant difference in preserving overall visual saliency.

Despite Scharr’s theoretical superiority in rotational symmetry and diagonal edge detection, the experiment proves that the structural integrity of the retargeted image is dictated by the global constraints of the dynamic programming pathfinding. Both operators suffer equally from semantic blindness and structural geometric artifacts ("kinks"), confirming that localized gradient energy is a fundamentally flawed proxy for high-level human perception.

To ensure seam carving remains viable in modern media pipelines, future work must transition away from purely mathematical energy estimators.

- **Semantic-Aware Seam Carving:** Integrate lightweight deep learning segmentation models (such as SegNeXt or YOLO-Seg) to dynamically mask primary subjects, guaranteeing that humans and critical objects are protected from anatomical distortion.
- **Forward Energy Optimization:** Implement Forward Energy criteria to explicitly penalize the creation of new artificial edges during the removal process, drastically reducing geometric bending in rigid architecture.
- **GPU-Accelerated Dynamic Programming:** Map the $\mathcal{O}(H \times W)$ dynamic programming accumulation matrix to a CUDA-based tensor pipeline to enable real-time retargeting for high-resolution media.
- **Spatio-Temporal Video Retargeting:** Extend the 2D energy algorithm into a 3D temporal volume using Graph Cuts to calculate a continuous "seam manifold," eliminating temporal jittering in video processing.

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